

CTA200 2026 Assignment 3: Figures

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Question 1

Figures 1, 2

Question 2

$$\dot{X} = -\sigma(X - Y) \quad (1)$$

$$\dot{Y} = rX - Y - XZ \quad (2)$$

$$\dot{Z} = -bZ + XY \quad (3)$$

$$\text{The Prandtl number : } \sigma = 10.0 \quad (4)$$

$$\text{The Rayleigh number : } r = 28.0 \quad (5)$$

$$\text{Dimensionless length parameter : } b = \frac{8}{3} \quad (6)$$

Figures 3, 4, 5, 6

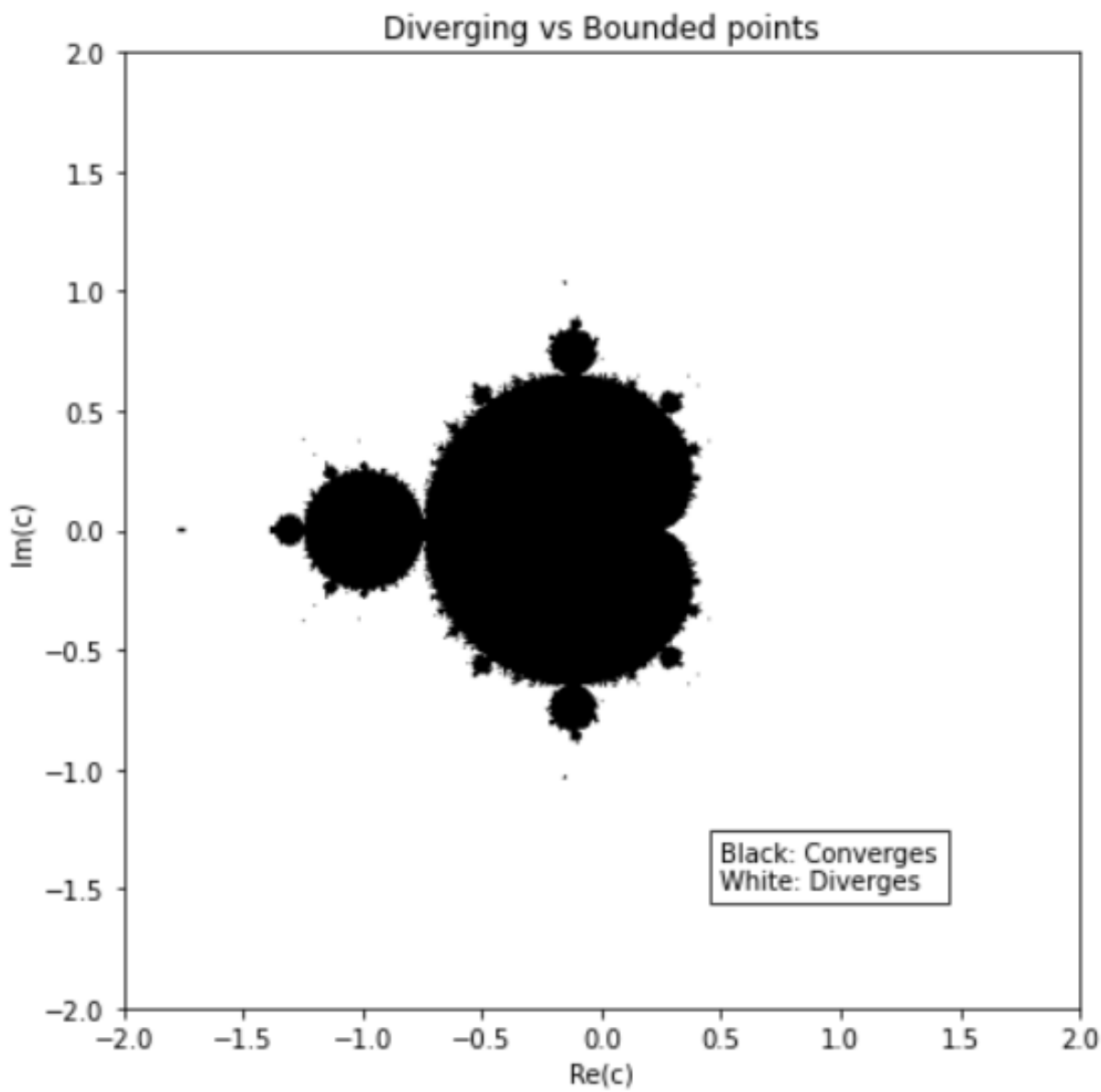


Figure 1: Figure depicting the convergence or divergence of $|z|$ with respect to c , where $z_{i+1} = z_i^2 + c$. Black indicates convergence, while white indicates divergence.

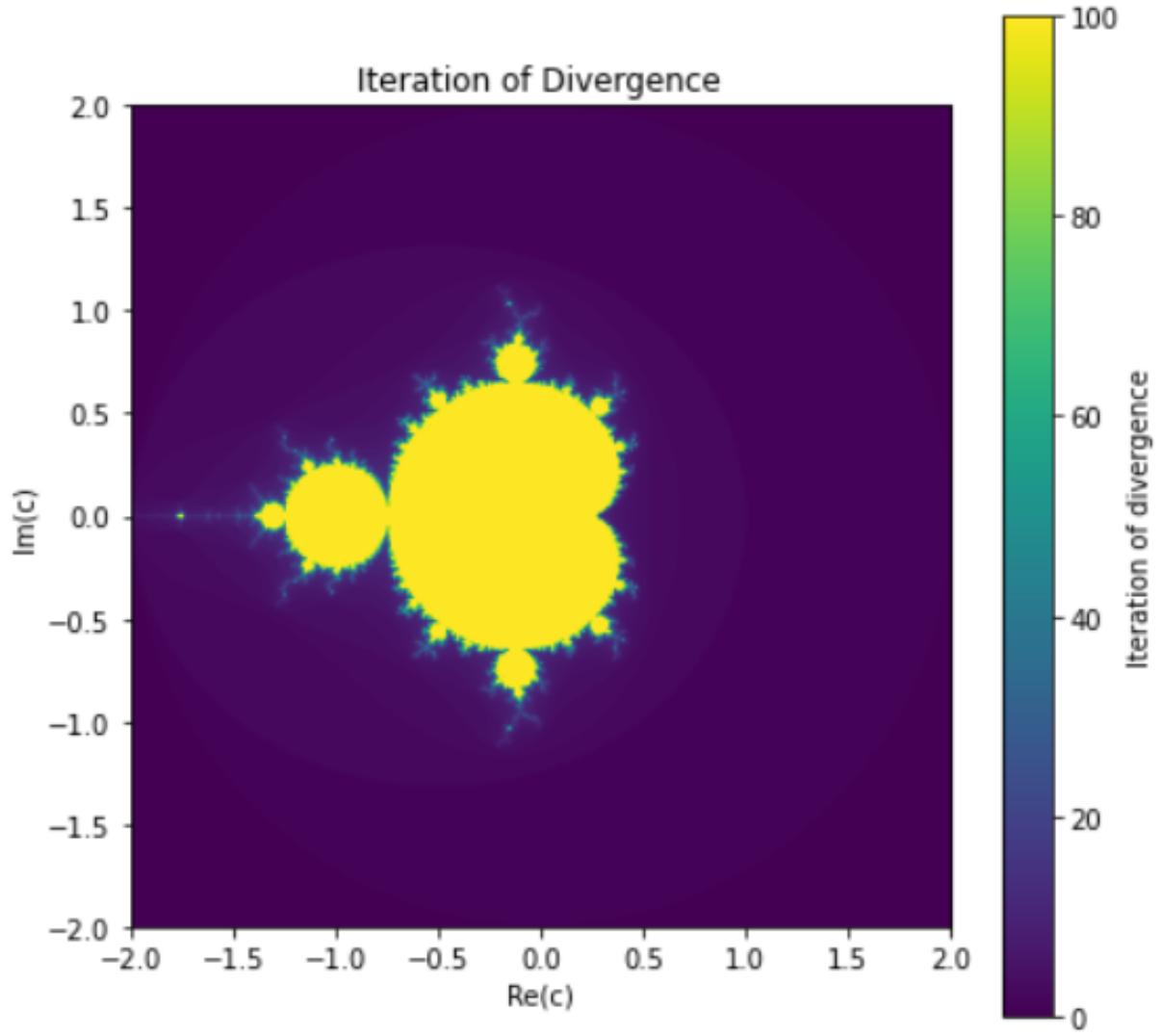


Figure 2: Figure depicting the convergence or divergence of $|z|$ with respect to c , where $z_{i+1} = z_i^2 + c$. Since the loop iterates 100 times, a value of yellow on the above figure indicates that the corresponding c does not lead to divergence.

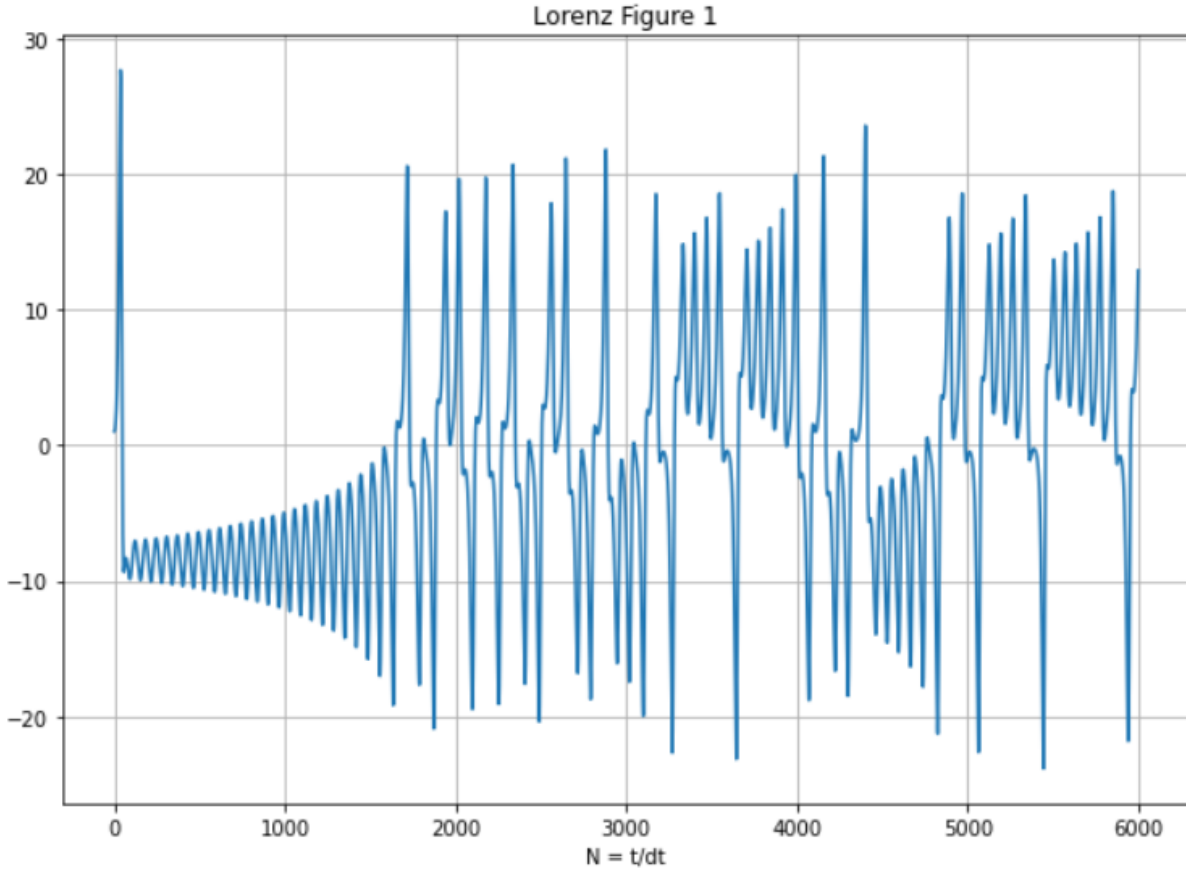


Figure 3: This figure displays the solution for Y plotted against $N = \frac{t}{\Delta t}$ where $\Delta t = 0.01$ over the time interval $t = 0$ to $t = 60$. The solution of Y was obtained through using `solve_ivp` on `.` This is a recreation of Lorenz' figure 1. It should be noted that Lorenz deconstructs his figure into 3 figures, of 1000 iterations each.

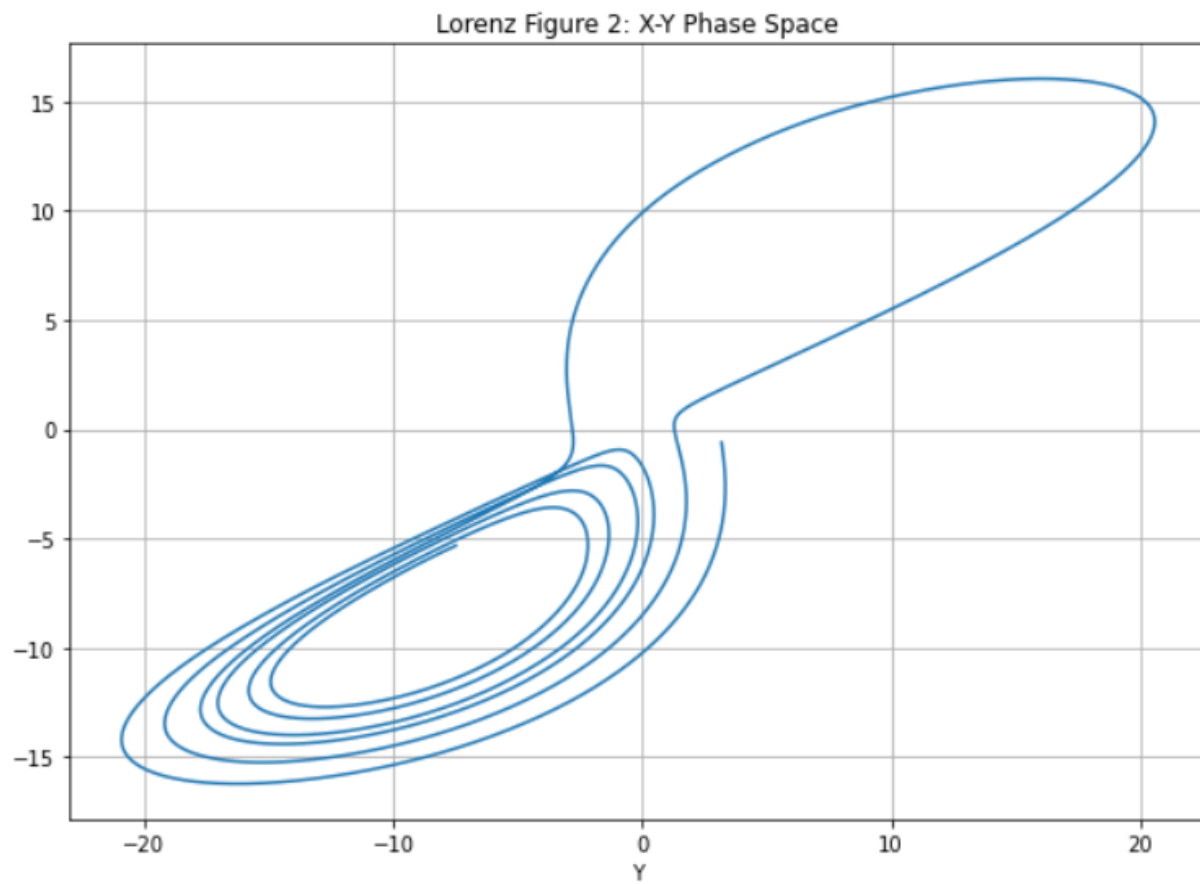


Figure 4: This figure depicts the X-Y plane in phase space, where the solutions to X and Y for times from $t = 14$ to $t = 19$ are obtained through solving equations using `solve_IVP`.

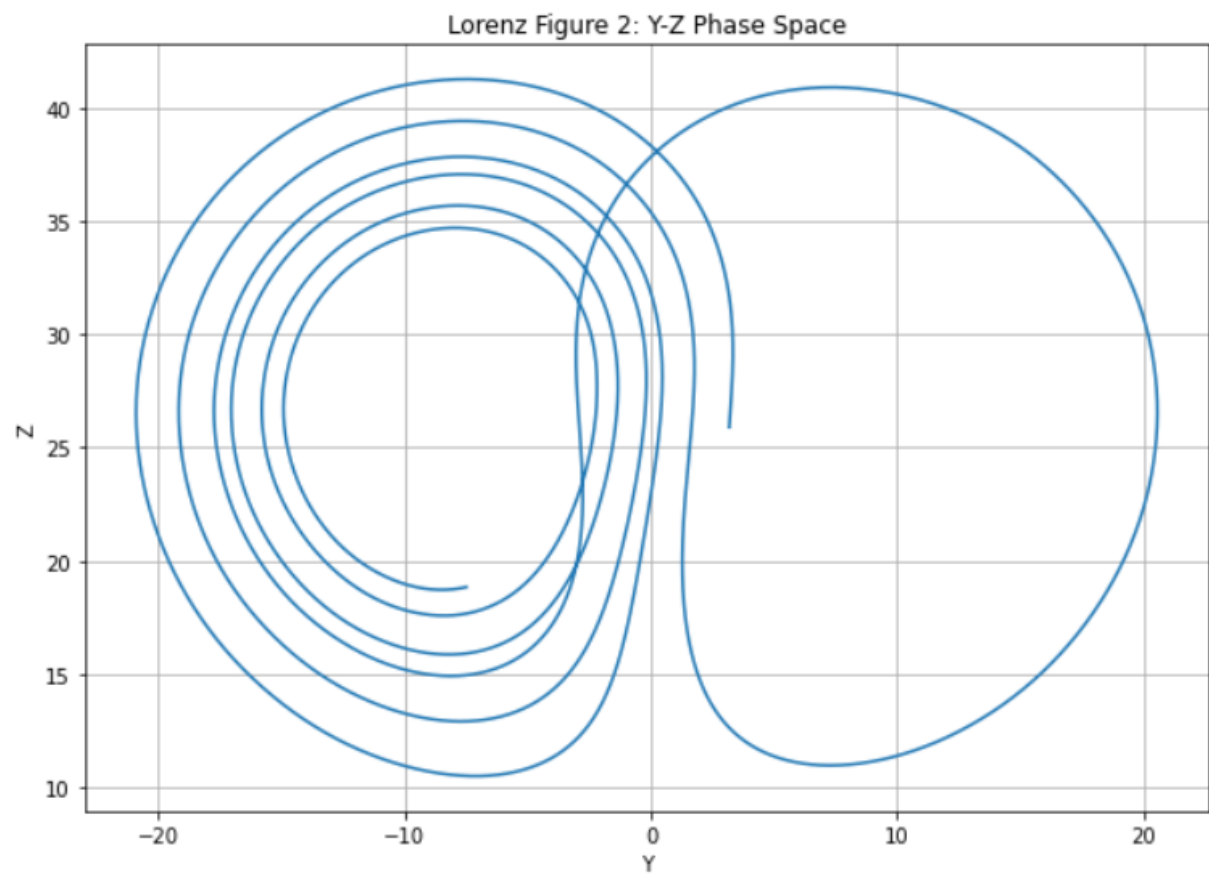


Figure 5: This figure depicts the Y-Z plane in phase space, where the solutions to X and Y for times from $t = 14$ to $t = 19$ are obtained through solving equations using `solve_IVP`.

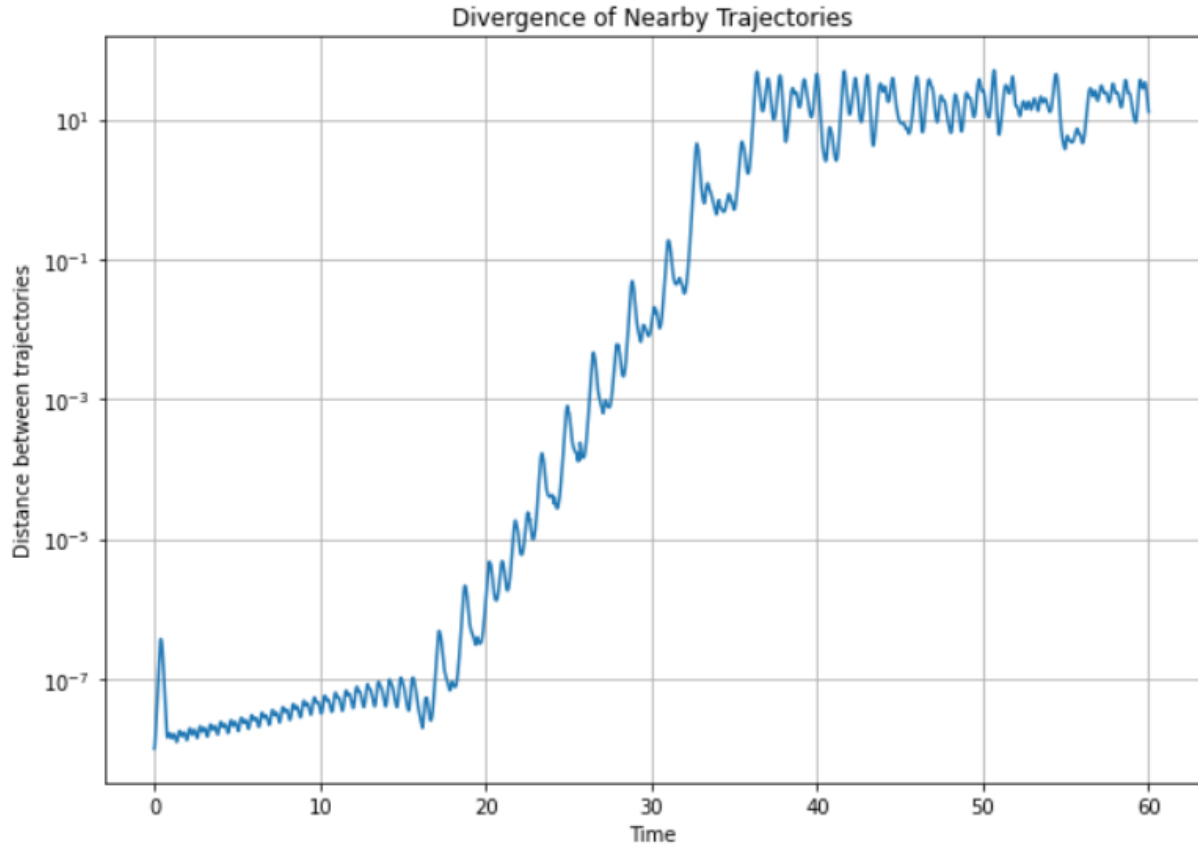


Figure 6: The distance between the solution trajectory predicted by initial conditions $W = [0., 1.0, 0.]$ and $W' = [0., 1.00000001, 0.]$. This is a semilog plot, with linear time and logged distance. A straight line indicates exponential behaviour.